



**Verified Carbon
Standard**

VCS VERIFICATION REPORT

INNER MONGOLIA WUDAOGOU 50.25MW WIND POWER PROJECT

(VCS PROJECT ID: 150)



Document Prepared By

Shenzhen CTI International Certification Co., Ltd

Project Title	Inner Mongolia Wudaogou 50.25MW Wind Power Project
Version	01.0
Report ID	CTI/NB-2022-0159-7

Report Title	VCS Verification Report of Inner Mongolia Wudaogou 50.25MW Wind Power Project
Client	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.

Pages	37
Date of Issue	20-July-2022
Prepared By	Shenzhen CTI International Certification Co., Ltd
Contact	Address: F8-A CTI Building, No.4 LiuXianSan Road, Xin'an Street, Bao'an District, 518101, Shen Zhen, China Tel: +86 10 65580012 Email: linshunrong@cti-cert.com Website: http://www.cti-cert.org
Approved By	Zhou Lu 
Work Carried Out By	Team Leader& Verifier: Feng Tian Technical Reviewer: Lin Shunrong

Summary:

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions reported for the project activity “Inner Mongolia Wudaogou 50.25MW Wind Power Project” (VCS Project ID:150) for the monitoring period 01/05/2012-31/12/2016 within the 1st VCS crediting period 01/01/2007-31/12/2016, to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity. These emission reductions are claimed as Verified Carbon Units (VCU) under the VCS Standard Version 4.3.

The verification was performed on the basis of VCS Standard Version 4.3 for the VCS projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspections, and the resolution of outstanding issues. The verification team identified one CL, and no CAR or FAR is raised in this monitoring period.

In CTI's opinion, the GHG emission reductions reported for the project in the monitoring report (version 02 dated 03/07/2022) are fairly stated. The GHG emission reductions of the period from 01/05/2012-31/12/2016 within the 1st VCS crediting period were calculated correctly on the basis of approved methodology ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” (version 6), the monitoring plan contained in the approved revised CDM-PDD (version 3.1, dated 14/08/2012).

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

Hence, CTI is able to certify that the emission reductions from the “Inner Mongolia Wudaogou 50.25MW Wind Power Project” during this monitoring period from 01/05/2012-31/12/2016 amount to 551,292 tCO₂e.

CONTENTS

1	Introduction	6
1.1	Objective	6
1.2	Scope and Criteria	6
1.3	Level of Assurance.....	7
1.4	Summary Description of the Project	7
2	Verification Process.....	8
2.1	Method and Criteria.....	8
2.2	Document Review	8
2.3	Interviews.....	9
2.4	Site Inspections.....	10
2.5	Resolution of Findings	11
2.6	Eligibility for Validation Activities	11
3	Validation Findings.....	12
3.1	Participation under Other GHG Programs	12
3.2	Methodology Deviations.....	12
3.3	Project Description Deviations.....	12
3.4	Grouped Project	13
4	Verification Findings.....	13
4.1	Project Implementation Status	13
4.2	Safeguards	20
4.3	AFOLU-Specific Safeguards	21
4.4	Accuracy of GHG Emission Reduction and Removal Calculations	21
4.5	Quality of Evidence to Determine GHG Emission Reductions and Removals	28
4.6	Non-Permanence Risk Analysis.....	28
5	Verification conclusion	28
	APPENDIX A: Abbreviations	31
	APPENDIX B: references	32
	APPENDIX C: Competence of team members and technical reviewers.....	35

APPENDIX D: Corrective Action Requests, clarification requests and Forward Action Requests.....	37
--	-----------

1 INTRODUCTION

Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd. has commissioned Shenzhen CTI International Certification Co., Ltd (CTI) to carry out the verification and certification of emission reductions reported for the “Inner Mongolia Wudaogou 50.25MW Wind Power Project” (the Project) for the monitoring period 01/05/2012-31/12/2016. This report contains the findings from the verification and includes a verification statement for the verified carbon units.

1.1 Objective

Verification is the periodic independent review and ex-post determination by an accredited verification body of the monitored reductions in GHG emissions that have occurred as a result of the registered VCS project activity during a defined verification period.

A verification statement is the written assurance by a verification body that, during a specific period in time, a project activity achieved the emission reductions as verified.

The objective of this verification was to verify and provide a verification statement of emission reductions reported for the “Inner Mongolia Wudaogou 50.25MW Wind Power Project” for the period 01/05/2012-31/12/2016.

1.2 Scope and Criteria

The scope of the verification is:

- To verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the approved revised monitoring plan;
- To evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement;
- To verify that reported GHG emissions data is sufficiently supported by evidence.

The criteria of the verification are:

- VCS Program Guide (version 4.2)/41/
- VCS Standard (version 4.3)/40/ and other relevant requirements defined by Verra;
- The approved methodology ACM0002 (version 6)/44/ applied by the project.

The verification shall ensure that reported emission reductions are complete and accurate in order to be verified.

1.3 Level of Assurance

The verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. As per the para 3.9.1 of the VCS Standard Version 4.3, the project size categorization belongs to 'Project' as the expected annual emission reductions is 142,848tCO₂e lower than 300,000 tCO₂e. Therefore, according to the requirement of para 4.1.8 (4) of VCS standard Version 4.3 "The threshold for materiality with respect to the aggregate of errors, omissions and misrepresentations relative to the total reported GHG emission reductions and/or removals shall be five percent for projects and one percent for large projects.", CTI applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities of the project activity.

1.4 Summary Description of the Project

Sectoral Scope and Project Type

According to the VCS Program Guide (version 4.2) /41/ and Annex A of the Kyoto Protocol, the project is applicable under the following activity categories:

- Sectoral Scope: 1. Energy (Renewable/non-renewable)
- Project Type: wind power project

Project Background

The project is located in Wengniute County, East Inner Mongolia, P.R.China. The site location's approximate coordinates are north latitude 42°39'-42°42' and east longitude 117°54'-118°01". The project activity was registered as a CDM project with Ref.1487 on 05/07/2008 and the renewable crediting period was selected starting on 05/07/2008. The project has been registered as VCS project with Ref. VCS 150 under VCS Standard Version 2007.1 with the crediting period start date 01/01/2007 when the project started full operation as confirmed by checking the Project completion report /33/, VCS validation report /23/, the VCS monitoring report and verification report for the period of 01/01/2007-30/06/2008 /27//28/.

The amount of 224,282 tCO₂e VCU's have been issued for the monitoring period 01/01/2007-30/06/2008 within the 1st VCS crediting period. GHG emission reductions have been applied for CERs under CDM of which total 522,728 tCO₂e CERs have been issued for the monitoring periods 05/07/2008-30/06/2009, 01/07/2009-30/06/2010, 01/07/2010-30/06/2011 and 01/07/2011-30/04/2012.

The installed capacity of the project is 50.25MW equipped with 67 sets of wind turbine generator in type of S48/750, with unit capacity of 750kW, manufactured by Xinjiang Gold Wind Science and Technology Co., Ltd. The electricity generated by the project activity is supplied to the Northeast China Power Grid

(hereafter referred to as “NEPG”), and the project is estimated to deliver 142,848 tCO₂e emission reductions annually.

2 VERIFICATION PROCESS

2.1 Method and Criteria

The verification was performed through means of the following three phases in accordance with the requirement of the approved revised CDM-PDD (version 3.1 dated 14/08/2012) and monitoring plan contained in the approved revised CDM-PDD, the applied methodology, and the VCS Standard (version 4.3) and other relevant VCS requirements:

- A desk review of the monitoring report and all support documents;
- Follow-up interviews with project stakeholders and site inspection;
- The resolution of outstanding issues and the issuance of the verification report and statement.

The following sections outline each step in more detail.

The verification of the emission reductions has assessed all factors and issues that constitute the basis for emission reductions from the project. These include:

- The emission reduction calculations and the relevant data records;
- The calibration and maintenance records for the monitoring instruments;
- The management systems to support the project operation and monitoring.

2.2 Document Review

Based on the requirements of competency, experience and qualified sectoral scopes, CTI appointed a verification team in accordance with CTI’s internal procedures.

Function	Name	Technical competence	Task Performance*
Team Leader& Verifier	Feng Tian	1.2	☒ DR ☒ SV ☒ RP ☐ TR
Technical Reviewer	Lin Shunrong	1.2, 14.1, 15.1	☒ DR ☐ SV ☐ RP ☒ TR

*DR=Document review; SV=Site visit; RP=Reporting; TR=Technical review

In addition to the VER/VCU monitoring report/1/, the approved CDM-PDD /26/, emission reduction calculation spreadsheet/2/during this monitoring period, the following documents also were assessed as a part of the verification audit:

- The VCS validation report /23/;

- The registered CDM-PDD and corresponding validation report /24/;
- The previous VER/VCU monitoring report/27/ and verification report/28/, and previous CDM/CER monitoring reports /29/and verification reports/30/;
- Baseline and monitoring methodology ACM0002 (version 6) applied by the project/44/;
- Relevant decisions, clarifications and guidance from the Verra/40/-/43/; and
- Other information and references relevant to the project activity.

During the desk review, CTI has applied standard auditing techniques to assess the quality of information provided. The following activities were performed:

- A review of the data and information presented to verify their completeness;
- A review of the approved revised monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures; and

An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

2.3 Interviews

On 23/05/2022, CTI visited the project proponent Chifeng Xinsheng Wind Power Co., Ltd. to perform on-site assessment. The key personnel of the project were interviewed or assisted the verification team /44/. Main topics of the interview cover implementation of the project construction, applicability of selected methodology, implementation of project monitoring, emission reduction calculation, etc.

The key personnel interviewed /45/ are summarized in the table below:

Interviewed personnel	Role	Organization	Subject
Mr. Wang Zhihai	Plant Director	Chifeng Xinsheng Wind Power Co., Ltd.	Operation of the project activity; Implementation of the monitor plan of the project activity; Data collection and data achievement; Calibration of meters and equipment maintenance; The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity. Project's sustainable development contributions
Mr. Wu Haibo	O&M Staff		
Mr. Liu Yiwei	O&M Staff		
Mr. Han Yulong	Local resident	Local Village	The impact of the project activity; The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity;
Mr. Lyu Zhijun			
Mr. Guo Shouqiang	Staff	Local Environment Protection Bureau	Project's sustainable development contributions。
Mr. Luo Sen	Project Manager	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.	Data collection and ER calculation.

2.4 Site Inspections

During the on-site assessment, CTI has applied standard auditing techniques to assess the quality of information provided. The following aspects of the project activity have been verified:

- An assessment of the implementation and operation of the registered project activity is as per the approved revised CDM PDD of the project activity;
- A review of information flows for generating, recording, aggregating and reporting the monitoring parameters; and
- Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the approved revised monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and the electricity sales receipts;
- A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the approved revised CDM PDD and the selected methodology;
- A review of calculations and assumptions made in determining the GHG data and emission reductions; and
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

In addition, all parameters required by the monitoring methodology ACM0002 (version 6), and the management system were assessed during the site visit.

2.5 Resolution of Findings

A corrective action request (CAR) shall be raised, where:

- i. Non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;
- ii. Mistakes have been made in applying assumptions, data or calculations of emission reductions which will impair the estimate of emission reductions;
- iii. Issues identified in a FAR during validation to be verified during verification have not been resolved by the project proponents.

A clarification request (CL) shall be raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

The verification team identified 1 CL and no CAR in this monitoring period. The CL was satisfactorily addressed by the project proponents in the updated monitoring report version 02 dated 03/07/2022 (refer to Appendix D for further details).

2.5.1 Forward Action Requests

A forward action request (FAR) is issued for actions if the monitoring and reporting require attention and/or adjustment for the next monitoring period.

CTI confirmed that there was no FAR identified in previous verification/28/, and no FAR was raised during this verification.

2.6 Eligibility for Validation Activities

The project has been validated by Det Norske Veritas under the CDM rules and registered as CDM project on 05/07/2008, and registered as VCS project under VCS standard version 2007.1 through gap validation by SGS United Kingdom Ltd.

CTI was accredited in Sectoral Scope 1 on validation and verification and thus is eligible for verification activities on the proposed project. By checking the UNFCCC website, the VCS pipeline and the previous verification reports/30/, it is confirmed the verification for this monitoring period 01/05/2012-31/12/2016 which is no more than consecutive 6 years is the first verification undertaken by CTI. Therefore, the verification undertaken by CTI is in line with the requirement for VVB stipulated in Section 4.1.20 of VCS standard version 4.3.

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

The project has been registered as a CDM project on 05/07/2008 under the UNFCCC with Ref. 1487 Th renewable CDM crediting period was selected starting on 05/07/2008. By checking the UNFCCC website: <https://cdm.unfccc.int/Projects/DB/DNV-CUK1198674466.99/view>, it is confirmed that total 522,728tCO₂e of CERs have been issued for the monitoring periods 05/07/2008-30/06/2009, 01/07/2009-30/06/2010, 01/07/2010-30/06/2011 and 01/07/2011-30/04/2012.

3.2 Methodology Deviations

The validation process /16/ has assessed all factors and issues that constitute the basis for emission reductions from the project according to the applicable CDM methodology ACM0002 (version 6) /44/. There was not any methodology deviation applied to this project. Details refer to section 4.1.

3.3 Project Description Deviations

By checking the previous VCS verification report with the monitoring period 01/01/2007-30/06/2008, the Project completion report /33/, and through the site visit interview with project owner, it is confirmed that the project started full operation on 01/01/2007 which is the project start date according to the requirement of VCS. A deviation was identified for the crediting period of the Project. The project is registered as VCS project under VCS version 2007.1 and completed validation before 19/03/2020. As per VCS requirement, it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice. Therefore, the first renewable crediting period of the Project under VCS is updated from 01/01/2007-30/06/2008 to 01/01/2007-31/12/2016, which lasts for 10 years. As per VCS requirement, the project did not conduct the validation of crediting period renewal within the two years after the end of the first VCS crediting period. Thus, the crediting period of the project ends on 31/12/2016.

CTI confirms that the project remains in compliance with VCS Program rules and this deviation has no impact on the applicability of the methodology, additionality or the appropriateness of the baseline scenario and meet all appropriate rules and requirements of VCS standard.

By checking the previous verification reports and the UNFCCC website: <https://cdm.unfccc.int/Projects/DB/DNV-CUK1198674466.99/view>, a revised monitoring plan has been approved by CDM EB on 22/07/2010 and the main changes include:

- The electricity exported to and imported from the grid through the main line are measured by the revenue meter (with accuracy degree of 0.2S) installed at the main transformer of the main line.
- The electricity imported from the grid through the 10kV backup line is monitored by the meter (with accuracy degree of 1.0) installed on the 10kV backup line.

In addition, a PRC has been approved by CDM EB on 27/12/2012 due to that the actual accuracy degree of the backup line meter is 2.0 which is lower than that stated in the revised monitoring plan approved by CDM EB on 22/07/2010. By checking the validation opinion on the PRC, the PRC is applied to the monitoring periods from 01/07/2011 onwards, therefore, the electricity imported from the grid through the 10kV backup line (EG_{backup}) is calculated as follow:

$$EG_{\text{backup}} = \text{measured value} + \text{measured value} \times 0.01.$$

The verification team confirmed that the deviations will not lower the accuracy and conservativeness of the monitoring system and will not negatively impact the conservativeness of the quantification of GHG emission reductions or removals. Furthermore, the deviation relates only to the criteria and procedures for monitoring or measurement, doesn't relate to any other part of the methodology.

In conclusion, the verification team assessed through visual inspection and document review that all physical features of the project activity including wind turbines, generators, transmission facilities, monitoring equipment, data monitoring, reporting and collecting systems have been implemented in accordance with the approved revised CDM PDD (version 3.1 dated 14/08/2012). CTI confirms that the project is implemented in accordance with the project description, and no any other project description deviations occurred. Details refer to section 4.1.

3.4 Grouped Project

The project was not a grouped project; hence this clause is not applicable.

4 VERIFICATION FINDINGS

This section summarizes the findings from the verification of the emission reductions reported for the "Inner Mongolia Wudaogou 50.25MW Wind Power Project" for the monitoring period 01/05/2012-31/12/2016.

4.1 Project Implementation Status

Project Implementation in accordance with the registered project design document

The project is a wind power grid-connected plant and locates Wengniute County, East Inner Mongolia, P.R.China. The geographical coordinates are verified by GPS system as north latitude 42°39'-42°42' and east longitude 117°54'-118°01". The total installed capacity is 50.25MW equipped with 67 sets of wind turbine generator in type of S48/750, with unit capacity of 750kW. It was confirmed by checking the operation logs/7/, the previous VCS/CDM monitoring reports and verification reports /27/-/30/ the first wind turbine was put into full operation on 01/01/2007. The actual implementation of the project during this verification period was verified in terms of nameplates of wind turbines /31/ and the equipment

purchase contract /34/. The details of the turbines with respect to their installation and capacity have been verified to be consistent with description indicated in the approved revised CDM PDD.

The electricity generated by the project activity is supplied to the NEPG, which can be confirmed by the Power purchase agreement (PPA) signed between Chifeng Xinsheng Wind Power Co., Ltd. and Grid Company/3/. All the monitoring system in operation period is consistent with the description in the monitoring plan of the approved PDD. The control system at the power plant is automated and assures continuous operation, including monitoring on malfunction of equipment. By checking the daily operation and maintenance records/7/ and via on-site visit, CTI can confirm that no serious malfunction happened and the plant was under a normal operation as expected in this monitoring period.

On-site training for the related procedures including, recording and reporting was verified to be in place/6/ and their implementation was confirmed by interview with the key operators/45/ and observing the operation.

By checking the VCS, UNFCCC website as well as other GHG programs and forms of credit such as exchange platform of Chinese Certified Emission Reduction (CCER) projects, Gold Standard, EU ETS, IREC, China ETS and via interviewing with the project proponent during on-site visit, it is confirmed that the emission reductions achieved during 01/07/2007-04/07/2008 have not been requested for issuance under any GHG programs. Via interviewing with the project proponent during on-site visit and checking the Clarification made by PP /36/, it is confirmed that PP voluntarily gives up the emission reductions during 01/07/2008-04/07/2008 and will not seek issuance under VCS or any other emission trading programs.

CTI confirms that the project implementation is in accordance with the approved revised CDM PDD. Furthermore, through visual inspection and document review, it is confirmed that all physical features of the project activity including data collection systems and storage systems have been implemented in accordance with the approved revised monitoring plan.

CL 01 was raised to specify the total number of years of the crediting period and has been successfully closed; details refer to Appendix D.

Compliance of monitoring plan with monitoring methodology

CTI is able to confirm that the monitoring plan contained in the approved revised CDM PDD is in accordance with the approved methodology applied by the project activity, i.e. ACM0002 (version 6).

Compliance of monitoring with the monitoring plan

CTI confirms that all parameters stated in the approved revised monitoring plan are monitored and reported appropriately. Parameters required to be monitored by the monitoring plan as per the monitoring methodology ACM0002 (version 6) and the management system were assessed during the site visit. The monitoring report lists each parameter required by the monitoring plan and the information flow (i.e. from data generation, recording, aggregation, calculation and reporting) for these parameters is provided. The information flow for each parameter is further verified in the following sections.

Parameters fixed ex ante

“Data and parameters fixed ex ante” in the MR are checked against the approved revised PDD. The parameter EF_y is the combined emission factor of the grid, which was determined ex-ante at the validation stage and has been fixed during the first crediting period. CTI verified and confirms that the emission factor used in the monitoring report is in compliance with the approved revised PDD.

Parameters monitored

CTI conducted document review and performed on-site assessment with project stakeholders to:

- A review of information flows for generating, aggregating and reporting the monitoring parameters;
- Determine whether the operational and data collection procedures are implemented in accordance with the approved revised monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and sales receipts;
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

According to the approved revised monitoring plan, the parameters to be monitored are as below:

EG_y Net electricity delivered to the grid by the project

Monitored parameter EG_y

EG_y is the net electricity delivered to the grid by the project and is calculated as the electricity exported to the grid (EG_{ex}) minus electricity imported from the grid through the main line (EG_{im}) and the 10kV backup line (EG_{backup}) which is as the equation below:

$$EG_y = EG_{ex} - EG_{im} - EG_{backup}$$

By site inspection and checking the monitoring plan contained in the approved revised CDM PDD and previous verification reports, CTI confirmed electricity exported to and imported from the grid through the main line are continuously measured by the revenue meter M1. The revenue meter M1 has been installed at the 35kV/220kV main transformer of the project and with an accuracy level of 0.2S.

Electricity imported from the grid through the 10kV backup line is continuously measured by meter M with an accuracy level of 2.0 installed on the 10kV backup line. According to the validation opinion on the revised PDD, as the accuracy of the backup line meter M is changed from 1.0 to 2.0, the measured values of electricity imported through the backup line is conservatively calculated as:

$$EG_{backup} = \text{measured value} + \text{measured value} \times 0.01.$$

The verification team had on-site checked the location of the meters against the diagram of power connection system /4/ and found them to be consistent and installed in accordance with the

approve revised monitoring plan. All the monitoring facilities and system have been verified by CTI during on-site visit.

By checking operation log /7/ and via on-site visit, it is confirmed that both meters were working well during this monitoring period. Via the on-site interview and by checking the daily operation and maintenance records/7/, monthly reading records and sales receipts/10/ issued by grid company, it is confirmed by the verification team that the electricity exported and imported through the main line are continuously measured by revenue meter M1 while electricity imported through the 10kV backup line is continuously measured by meter M. Readings of meters are monthly recorded by the designated personnel of the project proponent. The cut-off time of electricity exported to the grid through the main line and electricity imported through 10kV backup line is at 24:00 on the last day of each month while the cut-off time of electricity imported from the grid through the main line is at 00:00 on one day from 1st to 10th of each month which is determined by the grid company (In particular, PP would record the readings at 00:00 on the 1st day i.e. 01-Jan of every year in order to settle the readings of electricity imported from the main line for the whole calendar year). After confirmed by both sides the sales receipts/10/ were issued by the grid company and have been used for cross-checking the records of electricity exports and imports.

As described above, via the on-site inspection CTI verifies the meters have been installed in accordance with monitoring plan, further by checking the location of the meters against the diagram of power connection system to confirm the meter's location to be consistent.

Data in the monthly reading records were used to the report, through a cross check with sales receipts. The conservative values were used for the monitoring report and ERs calculation spreadsheet, which has been verified by the verification team. Supporting references and data required to determine the quantity of net electricity generation supplied by the project is found to be complete and transparent.

Monitoring equipment and calibration

The documents review was carried out. Particular attention was paid to the frequency of measurements, the quality of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the approved revised CDM PDD and the methodology ACM0002(version 6) and corresponding tool(s), and the quality assurance and quality control procedures.

All monitoring meters have been calibrated annually as per the approved revised monitoring plan covering this monitoring period. The relevant information of the calibration is listed as below:

Item	M1 (Revenue meter)	M (backup line meter)
Type	ZMD402	DTS8666
Accuracy	0.2S	2.0
Serial No.	86693424	00605518
Calibration frequency	Annually	

Calibration date	10/02/2012
	08/02/2013
	06/02/2014
	03/02/2015
	01/02/2016
Calibration validity	09/02/2013
	07/02/2014
	05/02/2015
	02/02/2016
	31/01/2017
Calibration entity	Northeast Electric Power Research Institute Co., Ltd.

Calibration records and accreditation certificates/12//13/have been verified by the verification team. The actual accuracy verified to be 0.2S for revenue meter M1 and 2.0 for backup line meter M which is consistent with the requirement in the monitoring plan contained in the approved revised CDM PDD and in compliance with the Technical administrative code of electric energy metering /15/. The calibration frequency of meters is annual, which is consistent with the approved revised monitoring plan. Based on the site visit and by checking the calibration reports, CTI confirms that:

- Both meters have been installed in accordance with the approved revised monitoring plan;
- The meters' accuracy is in line with the requirement of the approved revised monitoring plan;
- The calibration frequency of the meters is annual, which is in line with the approved revised monitoring plan and national technical standard /15/ as required by VCS standard. And the calibrations are verified to be valid for the whole reporting period.

Data management and control

Chifeng Xinsheng Wind Power Co., Ltd. is responsible for operation and routine maintenance of power plant under the project activity. The quality assurance and quality control procedures have been addressed in the project management and monitoring manual/5/, including the organization structure with the responsibilities, personnel competencies, monitoring procedures and monitoring management. By interview with the staff/45/ and check records /7//8/ during on-site visit, it can be confirmed that the monitoring management system is implemented following the project management and monitoring manual.

All monitoring devices have been calibrated and maintained periodically to ensure the accuracy of measurement. Calibration records of instruments used in measurements were made available during the verification visit and found to be valid for the entire period of the verification. Competence and training records of in-plant personnel engaged in measurement of plant parameters were presented during verification and found to be in order.

As the above and document review, verification team considers that the project activity operated properly in the way complying with the implementation plan and monitoring plan contained in the approved revised CDM PDD and the monitoring was performed in a transparent and comprehensive manner during this monitoring period.

By checking the UNFCCC website: <https://cdm.unfccc.int/Projects/DB/DNV-CUK1198674466.99/view>, it is confirmed that emission reductions within the monitoring periods 05/07/2008-30/06/2009, 01/07/2009-30/06/2010, 01/07/2010-30/06/2011 and 01/07/2011-30/04/2012 have been issued as CERs and no other issuance request was submitted under CDM.

Therefore, it is concluded no double counting issuance has occurred for this verified monitoring period 01/05/2012-31/12/2016 under both VCS and CDM programmes. Via the on-site interview and checking the Statement on no double counting made by PP /35/, it is confirmed that the emission reductions for this verified period are only requested for VCU's issuance.

The verification team has also checked other GHGs programs and forms of credit such as exchange platform of Chinese Certified Emission Reduction (CCER) projects, Gold Standard, EU ETS, IREC, China ETS and based on the interview with the project proponent, during this monitoring period, it is confirmed that except CDM and VCS scheme:

- No evidence showing that the project has participated or been rejected under any other GHG programs since validation or previous verification.
- There is no observation that the project has received or sought any other emission trading program and form of environmental credit, or has become eligible to do so since validation or previous verification.

The project is completely voluntary CDM and VCS project, not included in an emissions trading program or any other mechanism that includes GHG allowance trading. Although the national emission trading scheme has been launched since 16/07/2021, it only covers the high-emission and energy intensive industries that emit at least 26,000 tonnes of CO₂e/year as per the regulation/19/ including thermal power generation, petrochemical, chemical, building materials, iron and steel, etc. By checking the business license/14/of the project proponent, it is confirmed that the PP is qualified in business scope of renewable energy investment, thus is not included the mandatory emission control scheme and there is no emission cap enforced for the PP by further checking the compulsive company list under China ETS in public information/20/. Moreover, it is confirmed by CTI the project is not registered as a CCER project in China, the emission reductions generated by the project activity are not eligible for transaction under China ETS as per the regulation/19/. Hence, it is confirmed that the emission reductions will not be double counted.

Via the on-site inspection, and through the interviews with the project proponent, local villagers and official from local environment bureau, and by checking the evidences provided by the project proponent, the verification team agrees on the following points regarding to the promotion of sustainability development by the project activity:

- **SDG 7, Indicator 7.2.1 Renewable energy share in the total final energy consumption.**

The project provides clean and renewable energy through making use of wind power, displacing fossil fuel power in the grid NEPG. During this monitoring period, 481,102.502 MWh of net electricity generated by wind energy was delivered to the NEPG. It is verified by checking the monthly reading records/8/ and via on-site inspection. This also contributes to increasing the share of clean energy consumption of China's "Action Plan for Carbon Dioxide Peaking Before 2030" /38/.

- **SDG 8, Indicator Numbers of decent work opportunities provided which has income higher than local/national average level**

It is verified by interviewing with project proponent, staff and local villagers and checking the Statement on the annual income of employees from 2012 to 2016 by PP /21/ and Per capita disposable income of residents in Inner Mongolia Autonomous Region in 2012 and 2016 /22/, that the project activity provides 22 long-term job opportunities for local residents including 6 positions for female and provides higher salary from 99,834 RMB/year in 2012 to 126,582 RMB/year in 2016 above the local average level of 23,150 RMB/year in 2012 and 24,127 RMB/year in 2016. Hence, the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable. This also contributes to one of the China's actions for promoting sustainable developing, "Increase labor force participation rate through implementation of the classification policy. Vigorously enforce the Law on Promotion of Employment"/37/.

- **SDG 13, Indicator Tonnes of greenhouse gas emissions avoided or removed.**

As verified by on-site inspection, and via checking the emission reduction calculation sheet/2/ and the monthly reading records/8/, 551,292 tCO_{2e} GHG emission reductions are generated by the project activity during this monitoring period. This contributes achieve one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas"/39/.

Meanwhile, by checking the approved VCS and CDM monitoring reports for previous monitoring periods, it is confirmed that the current project contributions with all impacts were not included in previously approved VCS monitoring reports or reported in Sustainable Development Contribution Reports, thus, all impacts during the previous monitoring periods from 10/01/2007 to 30/04/2012 have not been calculated in the cumulative impact. Therefore, the cumulative contributions over the project lifetime are considered as the same as the current project contributions.

In conclusion, Verification Team was able to confirm that the project implementation is in accordance with the approved revised CDM PDD. All required equipments and procedures are available and implemented in an appropriate manner. All necessary monitoring instruments are installed and operated. The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on main meters have been observed during this monitoring period.

4.2 Safeguards

No Net Harm

Via on-site inspection and checking the EIA summary and conclusion provided in the approved revised CDM PDD, it is confirmed that the impact caused by the project activity on the surrounding ecosystem and residents, wastewater, noise, solid waste and atmosphere etc. is very little, there would be no net harm caused due to the project activity. Appropriate mitigation measures have been undertaken by the project proponent to avoid the potential impacts during the construction and operation. The EIA of the project are approved by the government.

Also, no potential environment or social economic matter was found during the site visit. It is verified by interviewing with project proponent, staff and local villagers and checking the Statement on the annual income of employees from 2012 to 2016 by PP /21/ and Per capita disposable income of residents in Inner Mongolia Autonomous Region in 2012 and 2016/22/, that the project has contributed to the local employment and improved income levels by providing job opportunities and offering permanent employees wages well above local average income levels. As stated by the plant staff during the on-site interviews, they have gained knowledge and skills in the regular training organized by the project proponent, so it can be judged that the project has, to some extent, improved the marketability of the employees as well as the possibility of lifelong learning for them.

In conclusion, the implementation of the project activity does not cause any significant negative environmental or socio-economic impacts and is beneficial to the local socio-economic environment.

Local Stakeholder Consultation

The local stakeholder's meeting was held in Wengniute County on 26/07/2006. Representatives from the local government and local residents attended the meeting. The project owner introduced the proposed project and the opinions expressed by the stakeholders were recorded and are available on request.

The stakeholder meeting showed that the proposed project receives strong support from the local community. They all believe the proposed project will promote local economic development and agree with the project development and construction.

For continuous communication with local stakeholders, the project owner has informed the local authority, the neighbourhood committee, and local residents of any key implementation schedules and changes of the project. The project owner also carried out questionnaire survey every two years to collect the relevant comments and suggestions, i.e. in June 2012, July 2014 and May 2016 during this monitoring period and no negative comment was received by checking the filled questionnaires through on-site visit. Through interviewing with staff from local environmental protection bureau and local residents during the site visit, it is confirmed by the verification team that during the implementation stage of this monitoring period, local authority has conducted spot checks on the implementation of the project periodically as per the request from the local governments' regulations, no negative comments

and issues from the local stakeholders during this monitoring period were identified and the project passed all the periodic spot checks by local government.

All such conclusion has been verified through site visit and checking approved revised CDM PDD.

4.3 AFOLU-Specific Safeguards

For non-AFOLU projects, this section is not required.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

CTI confirms that appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been followed, and the assumptions, emission factors and default values that are applied in the calculation have been justified.

According to the applied methodology, the emission reductions are determined as the difference between the baseline emissions, project emissions and leakage:

$$ER_y = BE_y - PE_y - L_y$$

Baseline emissions

Baseline emissions are determined as multiplying quantity of net electricity generation supplied by the project plant to the grid (EG_y) by the validated ex-ante fixed grid emission factor (EF_y).

$$BE_y = EG_y \times EF_y$$

Grid emission factor (EF_y)

EF_y is the grid emission factor of the which has been verified ex-ante in the validation stage in the approved revised CDM PDD as 1.1459 tCO₂e/MWh.

The net electricity supplied to the grid by the project (EG_y)

As per the Section 4.1 above, EG_y is determined the as electricity exported the grid (EG_{ex}) minus the electricity imported from the grid through the main line (EG_{im}) and 10kV backup line (EG_{backup}), i.e. $EG_y = EG_{ex} - EG_{im} - EG_{backup}$

The verification team has checked the monthly reading records, sales receipts issued by the grid company and emission reduction calculation sheet, and confirmed the data and calculation applied are correct, complete and transparent. The reading records and EG_y calculation are summarized in the Tables as follow:

Table 1 Electricity exported through the main line (EG_{ex}) and electricity imported from 10kV backup line (EG_{backup})

Period		EG_{ex} (MWh)			EG_{backup} (MWh)		
		Monthly Readings from M1	Values from sales receipts	Conservative Quantity	Monthly readings from meter M	Values from sales receipts	Conservative Quantity
From	To	A	B	$C=\min(A,B)$	G	H	$I=1.01*\max(G,H)$
01/05/2012	31/05/2012	9,701.832	9,701.832	9,701.832	0.000	0.000	0.000
01/06/2012	30/06/2012	6,810.000	6,810.000	6,810.000	0.720	0.720	0.727
01/07/2012	31/07/2012	4,850.000	4,850.000	4,850.000	0.000	0.000	0.000
01/08/2012	31/08/2012	3,327.296	3,327.296	3,327.296	0.840	0.840	0.848
01/09/2012	30/09/2012	3,816.935	3,816.935	3,816.935	0.000	0.000	0.000
01/10/2012	31/10/2012	11,715.578	11,715.578	11,715.578	0.000	0.000	0.000
01/11/2012	30/11/2012	13,195.766	13,195.766	13,195.766	0.000	0.000	0.000
01/12/2012	31/12/2012	12,548.489	12,548.489	12,548.489	0.000	0.000	0.000
2012 Subtotal		-	-	65,965.896	-	-	1.576
01/01/2013	31/01/2013	13,378.072	13,378.072	13,378.072	0.000	0.000	0.000
01/02/2013	28/02/2013	14,552.025	14,552.025	14,552.025	0.000	0.000	0.000
01/03/2013	31/03/2013	8,747.404	8,747.404	8,747.404	0.000	0.000	0.000
01/04/2013	30/04/2013	11,692.875	11,692.875	11,692.875	0.000	0.000	0.000
01/05/2013	31/05/2013	8,228.554	8,228.554	8,228.554	1.950	1.950	1.970
01/06/2013	30/06/2013	5,663.662	5,663.662	5,663.662	0.000	0.000	0.000
01/07/2013	31/07/2013	5,393.200	5,393.200	5,393.200	0.000	0.000	0.000
01/08/2013	31/08/2013	7,280.570	7,280.570	7,280.570	0.000	0.000	0.000
01/09/2013	30/09/2013	7,756.154	7,756.154	7,756.154	0.000	0.000	0.000
01/10/2013	31/10/2013	10,315.839	10,315.839	10,315.839	1.250	1.250	1.263
01/11/2013	30/11/2013	14,240.272	14,240.272	14,240.272	0.000	0.000	0.000
01/12/2013	31/12/2013	11,495.251	11,495.251	11,495.251	0.000	0.000	0.000
2013 Subtotal		-	-	118,743.878	-	-	3.232
01/01/2014	31/01/2014	15,286.382	15,286.382	15,286.382	1.330	1.330	1.343
01/02/2014	28/02/2014	5,323.896	5,323.896	5,323.896	0.000	0.000	0.000
01/03/2014	31/03/2014	7,944.352	7,944.352	7,944.352	0.000	0.000	0.000

01/04/2014	30/04/2014	7,449.578	7,449.578	7,449.578	0.000	0.000	0.000
01/05/2014	31/05/2014	11,772.706	11,772.706	11,772.706	0.000	0.000	0.000
01/06/2014	30/06/2014	2,856.188	2,856.188	2,856.188	0.000	0.000	0.000
01/07/2014	31/07/2014	6,218.833	6,218.833	6,218.833	0.780	0.780	0.788
01/08/2014	31/08/2014	2,202.992	2,202.992	2,202.992	0.000	0.000	0.000
01/09/2014	30/09/2014	4,795.591	4,795.591	4,795.591	0.000	0.000	0.000
01/10/2014	31/10/2014	7,958.828	7,958.828	7,958.828	0.000	0.000	0.000
01/11/2014	30/11/2014	12,850.234	12,850.234	12,850.234	0.000	0.000	0.000
01/12/2014	31/12/2014	15,777.479	15,777.479	15,777.479	0.000	0.000	0.000
2014 Subtotal		-	-	100,437.059	-	-	2.131
01/01/2015	31/01/2015	11,655.849	11,655.849	11,655.849	0.000	0.000	0.000
01/02/2015	28/02/2015	7,547.157	7,547.157	7,547.157	0.000	0.000	0.000
01/03/2015	31/03/2015	12,055.975	12,055.975	12,055.975	0.000	0.000	0.000
01/04/2015	30/04/2015	10,481.015	10,481.015	10,481.015	0.000	0.000	0.000
01/05/2015	31/05/2015	11,563.948	11,563.948	11,563.948	0.000	0.000	0.000
01/06/2015	30/06/2015	4,735.701	4,735.701	4,735.701	0.670	0.670	0.677
01/07/2015	31/07/2015	1,926.892	1,926.892	1,926.892	0.000	0.000	0.000
01/08/2015	31/08/2015	3,080.514	3,080.514	3,080.514	0.000	0.000	0.000
01/09/2015	30/09/2015	4,784.169	4,784.169	4,784.169	0.000	0.000	0.000
01/10/2015	31/10/2015	11,033.792	11,033.792	11,033.792	0.571	0.571	0.577
01/11/2015	30/11/2015	6,840.015	6,840.015	6,840.015	0.000	0.000	0.000
01/12/2015	31/12/2015	10,148.018	10,148.018	10,148.018	0.000	0.000	0.000
2015 Subtotal		-	-	95,853.045	-	-	1.253
01/01/2016	31/01/2016	6,079.787	6,079.787	6,079.787	0.000	0.000	0.000
01/02/2016	29/02/2016	10,333.084	10,333.084	10,333.084	0.230	0.230	0.232
01/03/2016	31/03/2016	11,201.073	11,201.073	11,201.073	0.000	0.000	0.000
01/04/2016	30/04/2016	10,226.520	10,226.520	10,226.520	0.000	0.000	0.000
01/05/2016	31/05/2016	13,308.354	13,308.354	13,308.354	0.000	0.000	0.000
01/06/2016	30/06/2016	4,397.406	4,397.406	4,397.406	0.000	0.000	0.000
01/07/2016	31/07/2016	2,629.447	2,629.447	2,629.447	0.000	0.000	0.000
01/08/2016	31/08/2016	2,878.811	2,878.811	2,878.811	0.000	0.000	0.000
01/09/2016	30/09/2016	3,997.546	3,997.546	3,997.546	0.000	0.000	0.000
01/10/2016	31/10/2016	10,086.042	10,086.042	10,086.042	0.980	0.980	0.990

01/11/2016	30/11/2016	15,786.432	15,786.432	15,786.432	0.000	0.000	0.000
01/12/2016	31/12/2016	12,904.051	12,904.051	12,904.051	0.000	0.000	0.000
2016 Subtotal		-	-	103,828.553	-	-	1.222
Total (01/05/2012-31/12/2016)		-	-	484,828.431	-	-	9.414

*Note:

During this monitoring period, the cut-off time for EG_{ex} and EG_{backup} is at 24:00 on the last day of each month, and sales receipts were issued by grid company for the cross check. Most conservative values are applied for baseline calculations.

Table 2 The electricity imported from the grid through the main line (EG_{im})

EG _{im} (MWh)				
Period*		Monthly readings from M1***	Values from sales receipts****	Conservative Quantity
From	To	D	E	F= Max (D, E)
04/04/2012	03/05/2012**	19.503	43.773	43.773
04/05/2012	04/06/2012	24.255	54.439	54.439
05/06/2012	04/07/2012	31.284	70.215	70.215
05/07/2012	04/08/2012	54.648	122.654	122.654
05/08/2012	03/09/2012	45.936	103.101	103.101
04/09/2012	04/10/2012	54.945	123.321	123.321
05/10/2012	04/11/2012	11.781	26.442	26.442
05/11/2012	03/12/2012	3.366	7.555	7.555
04/12/2012	31/12/2012	11.979	26.886	26.886
2012 Subtotal		-	-	534.613
01/01/2013	03/02/2013	14.454	32.441	32.441
04/02/2013	04/03/2013	10.098	22.664	22.664
05/03/2013	02/04/2013	11.583	25.997	25.997
03/04/2013	05/05/2013	45.540	102.212	102.212
06/05/2013	03/06/2013	20.493	45.995	45.995
04/06/2013	03/07/2013	53.262	119.544	119.544
04/07/2013	03/08/2013	49.599	111.322	111.322
04/08/2013	24/09/2013	16.830	37.774	37.774
25/09/2013	04/10/2013	7.920	17.776	17.776
05/10/2013	07/11/2013	22.671	50.884	50.884

08/11/2013	09/12/2013	6.633	14.887	14.887
10/12/2013	31/12/2013	17.820	39.996	39.996
2013 Subtotal		-	-	621.492
01/01/2014	08/02/2014	16.929	37.996	37.996
09/02/2014	09/03/2014	36.531	81.992	81.992
10/03/2014	08/04/2014	37.521	84.214	84.214
09/04/2014	08/05/2014	31.779	71.326	71.326
09/05/2014	08/06/2014	22.770	51.106	51.106
09/06/2014	09/07/2014	53.460	119.988	119.988
10/07/2014	07/08/2014	38.610	86.658	86.658
08/08/2014	08/09/2014	49.500	111.100	111.100
09/09/2014	08/10/2014	49.500	111.100	111.100
09/10/2014	08/11/2014	25.740	57.772	57.772
09/11/2014	08/12/2014	5.940	13.332	13.332
09/12/2014	31/12/2014	7.920	17.776	17.776
2014 Subtotal		-	-	844.360
01/01/2015	08/02/2015	25.740	57.772	57.772
09/02/2015	08/03/2015	14.850	33.330	33.330
09/03/2015	08/04/2015	19.800	44.440	44.440
09/04/2015	08/05/2015	25.740	57.772	57.772
09/05/2015	08/06/2015	13.860	31.108	31.108
09/06/2015	08/07/2015	53.460	119.988	119.988
09/07/2015	08/08/2015	54.450	122.210	122.210
09/08/2015	08/09/2015	87.120	195.536	195.536
09/09/2015	08/10/2015	32.670	73.326	73.326
09/10/2015	08/11/2015	34.650	77.770	77.770
09/11/2015	08/12/2015	36.630	82.214	82.214
09/12/2015	31/12/2015	14.751	33.108	33.108
2015 Subtotal		-	-	928.574
01/01/2016	08/01/2016	0.000	11.332	11.332
09/01/2016	31/01/2016	19.899	33.330	33.330

01/02/2016	06/03/2016	27.819	62.438	62.438
07/03/2016	06/04/2016	32.175	72.215	72.215
07/04/2016	06/05/2016	43.956	98.657	98.657
07/05/2016	06/06/2016	20.889	46.884	46.884
07/06/2016	06/07/2016	51.381	115.322	115.322
07/07/2016	06/08/2016	32.076	71.993	71.993
07/08/2016	06/09/2016	24.354	54.661	54.661
07/09/2016	06/10/2016	50.490	113.322	113.322
07/10/2016	06/11/2016	30.294	67.993	67.993
07/11/2016	06/12/2016	17.523	39.329	39.329
07/12/2016	31/12/2016	4.059	9.110	9.110
2016 Subtotal		-	-	787.476
Total (01/05/2012-31/12/2016)		-	-	3,716.515

Note:

* During this monitoring period, the cut-off time for EG_{im} is at 00:00 of one day from 1st to 10th of each month determined by the grid company (In particular, the grid company would record the readings at 00:00 on the 1st day i.e. 01-Jan of every year in order to settle the readings for the whole calendar year) and sales receipts were issued by grid company for the cross check. Most conservative values are applied for baseline calculations.

**As EG_{im} for 01/05/2012 to 03/05/2012 was settled in the reading records and sales receipts for 04/04/2012 to 03/05/2012 and there is no separate records and receipts for those three days. To be conservative, the EG_{im} for the whole period of 04/04/2012 to 03/05/2012 was included in the emission calculation.

*** The monthly readings of EG_{im} are taken from the monthly reading records /8/ and the verification team has checked the readings with the original copy of the monthly readings /9/ and found them to be consistent.

****The sales receipts of EG_{im} are issued together with another VCS project (VCS ID 395) and the receipts only display the sum values of the two projects. To be conservative, the sum values of the two projects from the sales receipts are used for ER calculations, CTI accepts this method as it is reasonable and conservative.

Table 3 Net electricity delivered to the grid by the project (EG_y)

Period	EG _{ex} (MWh)	EG _{im} (MWh)	EG _{backup} (MWh)	EG _y (MWh)
-	C	F	I	J=C-F-I
01/05/2012-31/12/2012	65,965.896	534.613	1.576	65,429.707
01/01/2013-31/12/2013	118,743.878	621.492	3.232	118,119.154
01/01/2014-31/12/2014	100,437.059	844.360	2.131	99,590.568
01/01/2015-31/12/2015	95,853.045	928.574	1.253	94,923.218
01/01/2016-31/12/2016	103,828.553	787.476	1.222	103,039.855
Total	484,828.431	3,716.515	9.414	481,102.502

The corresponding baseline emissions during this monitoring period is calculated as:

Year	EG _y (MWh)	EF _y (tCO ₂ /MWh)	BE _y (tCO ₂)
2012 (01/05/2012-31/12/2012)	65,429.707	1.1459	74,975
2013	118,119.154	1.1459	135,352
2014	99,590.568	1.1459	114,120
2015	94,923.218	1.1459	108,772
2016	103,039.855	1.1459	118,073
Total	481,102.502	-	551,292

Project emissions

As statement in the approved revised CDM PDD, for the wind power activities, the project emissions from the project is zero. Hence, PE_y during the monitoring period from **Inner Mongolia Wudaogou 50.25MW Wind Power Project** is zero.

Leakages

Leakage does not need to be accounted for this project as per the monitoring methodology ACM0002 (version 6).

Emission reductions

The emission reductions for this monitoring period were calculated as:

Monitoring period	GHG emission reductions or removals (tCO ₂ e)
01/05/2012 – 31/12/2012	74,975
01/01/2013 – 31/12/2013	135,352
01/01/2014 – 31/12/2014	114,120
01/01/2015 – 31/12/2015	108,772
01/01/2016 – 31/12/2016	118,073
Total ERs claimed (in 1,706 days)	551,292

Comparison of actual emission reductions or net anthropogenic GHG removals by sinks with estimates in approved revised CDM PDD

The emission reductions claimed are 551,292 tCO₂e in this monitoring period (i.e. 1,706 days). Compared with expected emission reductions 667,668 tCO₂e (calculated as $142,848/365d \times 1,706d$), the reported emission reductions in this monitoring period which are 17.43% lower than that expected in the approved revised CDM PDD. Considering the natural fluctuate of the wind resource, and actual generator operation conditions, CTI is able to confirm that the actual power supply and also emission reductions reported in this monitoring period are reasonable and appropriate.

CTI verified the input data for calculating emission reductions and the calculating process, and confirmed the result were complete and transparent. Therefore, CTI is able to confirm that the actual power supply and also emission reductions reported in this monitoring period are reasonable and appropriate.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

All necessary documentations are collected, referenced and aggregated, which is easily accessible in hard-copy or electronic format. Measurements are performed by calibrated equipment, and the key data can also be cross-checked via other sources, such as records, receipts and inventory data. No assumptions are used that have any material influence on reported emission reductions.

CTI concludes that during this monitoring period, the evidences for determination of emission reductions are sufficient and reasonable, and the calculation of emission reductions is reliable.

4.6 Non-Permanence Risk Analysis

The project is not AFOLU project, and thus non-permanence risk analysis is not applicable for the project.

5 VERIFICATION CONCLUSION

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions that have been reported for the project activity “Inner Mongolia Wudaogou 50.25MW Wind Power Project” in China (VCS Project ID: 150) for the period 01/05/2012-31/12/2016.

The verification is based on the baseline and monitoring methodology ACM0002 (version 6), the approved revised CDM PDD. The verification consisted of the following three phases: i) desk review of the project design, the baseline and approved revised monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and the issuance of the final verification and certification report.

The project proponents are responsible for the collection, calculation and determination of the GHG data in accordance with the approved revised monitoring plan and the reporting of GHG emission reductions on the basis set out within the project monitoring report.

Our verification approach was based on the requirements as defined under the applicable VCS Standard version 4.3 and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the project is implemented and operated as per the approved revised CDM PDD ;
- the revised monitoring plan is as per the applied methodology;
- the monitoring complies with the approved revised monitoring plan;
- the monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS Standard version 4.3 and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.

It is CTI's responsibility to provide an independent verification statement on the reported GHG emission reductions for the project. Based on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these, CTI planned and performed our work to obtain the information and explanations that we considered necessary to provide reasonable assurance that reported GHG emission reductions are fairly stated.

CTI does not assume any responsibility towards the issuance and utilization of the VCU's hereby verified and certified. Request for issuance of VCU's shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

In CTI's opinion the GHG emissions reductions of the "Inner Mongolia Wudaogou 50.25MW Wind Power Project" for the period 01/05/2012-31/12/2016 are fairly stated in the monitoring report (version 02 dated 03/07/2022). The GHG emission reductions were calculated correctly on the basis of the approved methodology ACM0002 (version 6) and the monitoring plan contained in the approved CDM PDD.

CTI can confirm that the GHG emission reductions are calculated without material misstatements. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CTI confirms the following statement:

Reporting period: 01/05/2012-31/12/2016

Verified GHG emission reductions and removals in the above verification period:

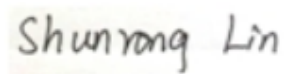
Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2012 (01/05/2012-31/12/2012)	74,975	0	0	74,975
2013	135,352	0	0	135,352
2014	114,120	0	0	114,120
2015	108,772	0	0	108,772
2016	118,073	0	0	118,073
Total	551,292	0	0	551,292



Mr. Feng Tian

Team Leader

20-July-2022



Ms. Lin Shunrong

Technical Reviewer

20-July-2022

APPENDIX A: ABBREVIATIONS

CAR	Corrective Action Request
CCER	Chinese Certified Emission Reduction
CER	Certified Emission Reduction(s)
CL	Clarification request
CO ₂	Carbon Dioxide
CO ₂ e	Carbon Dioxide Equivalent
CTI	Shenzhen CTI International Certification Co., Ltd
DOE	Designated Operational Entity
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reduction
ETN	Electricity Transaction Note
EU ETS	EU Emissions Trading Scheme
FAR	Forward Action Request
FSR	Feasibility Study Report
GHG	Greenhouse Gas(es)
IREC	International Renewable Energy Certificates
MR	Monitoring Report
NEPG	Northeast China Power Grid
PD	Project Description
PP	Project Proponent
SN	Serial Number
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit

APPENDIX B: REFERENCES

Documentation used to verify the information provided by the project proponents

/1/	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.: VER/VCU Monitoring Report for Inner Mongolia Wudaogou 50.25MW Wind Power Project, version 01 dated 10/05/2022, version 02 dated 03/07/2022
/2/	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.: Emission reduction calculation spreadsheet for Inner Mongolia Wudaogou 50.25MW Wind Power Project, version 02 dated 28/06/2022
/3/	Chifeng Xinsheng Wind Power Co., Ltd. and Grid Company: Power Purchase Agreement for Inner Mongolia Wudaogou 50.25MW Wind Power Project
/4/	Chifeng Xinsheng Wind Power Co., Ltd.: Diagram of power connection system
/5/	Chifeng Xinsheng Wind Power Co., Ltd.: VER monitoring manual and management procedure.
/6/	Chifeng Xinsheng Wind Power Co., Ltd.: Records of training for on-site staff.
/7/	Chifeng Xinsheng Wind Power Co., Ltd.: Daily operation and maintenance records.
/8/	Chifeng Xinsheng Wind Power Co., Ltd.: Monthly reading records of Inner Mongolia Wudaogou 50.25MW Wind Power Project from 01/05/2012-31/12/2016
/9/	Chifeng Xinsheng Wind Power Co., Ltd.: Original copy of the monthly readings of Inner Mongolia Wudaogou 50.25MW Wind Power Project from 01/05/2012-31/12/2016
/10/	Grid Company.: Electricity Sales Receipts from 01/05/2012-31/12/2016
/11/	Questionnaires filled by local stakeholders
/12/	China National Accreditation Service for Conformity Assessment: Accreditation certificates of Northeast Electric Power Research Institute Co., Ltd., issued on 15/01/2012 and valid to 14/01/2017
/13/	Northeast Electric Power Research Institute Co., Ltd.: Calibration certificates for revenue meter M1 and backup line meter M covering this monitoring period
/14/	Business license of Chifeng Xinsheng Wind Power Co., Ltd.
/15/	State Economic and Trade Commission: Technical administrative code of electric energy metering (DL/T 448-2016)
/16/	UNFCCC, http://cdm.unfccc.int

/17/	Gold Standard, http://www.goldstandard.org/
/18/	China CCER Exchange Info-platform, http://cdm.ccchina.org.cn/ccer.aspx
/19/	Measures for the Administration of Carbon Emission Trading, issued by Ministry of Ecology and Environment of China, dated 31/12/2020
/20/	Compulsive company list in China cap & trade scheme, issued by Ministry of Ecology and Environment of China
/21/	Chifeng Xinsheng Wind Power Co., Ltd.: Statement on the annual income of employees from 2012 to 2016
/22/	Per capita disposable income of residents in Inner Mongolia Autonomous Region in 2012 and 2016 https://www.nmg.gov.cn
/23/	SGS United Kingdom Ltd: VCS Validation Report
/24/	Registered CDM-PDD, version 3.0 dated 14/11/2007 Det Norske Veritas: CDM Validation Report, version 02, dated 21/12/2007
/25/	Revised monitoring plan approved by EB on 22/07/2010 Validation Report of revised monitoring plan
/26/	Approved CDM-PDD, version 3.1 dated 14/08/2012 Bureau Veritas Certification Holding SAS: Validation report, version 01 dated 07/09/2012
/27/	VER/VCU Monitoring Report for Inner Mongolia Wudaogou 50.25MW Wind Power Project of previous monitoring period 01/01/2007-30/06/2008
/28/	SGS United Kingdom Ltd: VCS Verification Report for Inner Mongolia Wudaogou 50.25MW Wind Power Project of previous monitoring period 01/01/2007-30/06/2008
/29/	CDM/CER Monitoring Reports for Inner Mongolia Wudaogou 50.25MW Wind Power Project of previous CDM monitoring periods 05/07/2008-30/06/2009, 01/07/2009-30/06/2010, 01/07/2010-30/06/2011 and 01/07/2011-30/04/2012
/30/	CDM Verification Reports for Inner Mongolia Wudaogou 50.25MW Wind Power Project of previous monitoring period 05/07/2008-30/06/2009, 01/07/2009-30/06/2010, 01/07/2010-30/06/2011 and 01/07/2011-30/04/2012
/31/	Xinjiang Gold Wind Science and Technology Co., Ltd.: Nameplates of the equipment
/32/	Feasibility Study Report of Inner Mongolia Wudaogou 50.25MW Wind Power Project
/33/	Chifeng Xinsheng Wind Power Co., Ltd.: Project completion report of the Project

/34/	Co-signed by the Xinjiang Gold Wind Science and Technology Co., Ltd., and Chifeng Xinsheng Wind Power Co., Ltd.: Wind Turbines Purchase Agreement
/35/	Chifeng Xinsheng Wind Power Co., Ltd.: Statement on no double counting
/36/	Chifeng Xinsheng Wind Power Co., Ltd.: Clarification on giving up the emission reductions during 01/07/2008-04/07/2008.
/37/	Law of the People's Republic of China on Promotion of Employment Order of the President of the People's Republic of China
/38/	Action Plan for Carbon Dioxide Peaking Before 2030 published by the State Council on 24/10/2021
/39/	Responding to Climate Change: China's Policies and Actions released by the State Council on 10/2021

Methodologies, tools and other guidance

/40/	Verified Carbon Standard: VCS Standard, version 4.3
/41/	Verified Carbon Standard: VCS Program Guide, version 4.2
/42/	Verified Carbon Standard: VCS Sectoral Scopes http://v-c-s.org/node/448
/43/	Verified Carbon Standard: Registration and Issuance Process, version 4.2
/44/	UNFCCC EB: Approved methodology, ACM0002 , version 6

Persons interviewed

	Interviewed personnel	Role	Organization
/45/	Mr. Wang Zhihai	Plant Director	Chifeng Xinsheng Wind Power Co., Ltd.
	Mr. Wu Haibo	O&M Staff	
	Mr. Liu Yiwei	O&M Staff	
	Mr. Han Yulong	Local resident	Local Village
	Mr. Lyu Zhijun		
	Mr. Guo Shouqiang	Staff	Local Environment Protection Bureau
	Mr. Luo Sen	Project Manager	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.

APPENDIX C: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Mr. Tian FENG

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification					
GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
√	-	√	√	-	-

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu LIN

Wu Lin

Technical Competent Manager

Shenzhen, 10/02/2022

Ms. Shunrong LIN

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 14: Afforestation and reforestation	TA 14.1: Afforestation and reforestation
SS 15: Agriculture	TA 15.1: Agriculture

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by: *Wu Lin*
Wu LIN

Technical Competent Manager

Shenzhen, 01/01/2021

APPENDIX D: CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS

Table 1: Corrective Action Requests

CAR ID	Corrective Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA

Table 2: Clarification Requests

CL ID	Clarification Request	Response by Project Proponent	Verification Team Assessment
CL 01	Total number of years of the crediting period is missing in MR (version 01 dated 10/05/2022) thus, CL 01 was raised for clarification	The total number of years had been included in the updated in the MR.	OK. It has been revised and verified by CTI by checking updated MR. Therefore, CL 01 is closed.

Table 3: Forward Action Requests

FAR ID	Forward Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA